

Chapter 114 — Conservative Management of Urinary Incontinence: Behavioural and Pelvic Floor Therapy, Urethral and Pelvic Devices

- **Conservative treatment** of urinary incontinence (UI) and associated lower urinary tract symptoms (LUTS) is any therapy that **does not involve surgery or pharmacotherapy**.
- Behavioural treatments are the **mainstay of conservative treatment** because they are effective and **essentially risk free**, and several guidelines and consensus panels recommend them as **first-line therapy**.
- **Definition:** behavioural treatments improve UI and LUTS by **changing the patient's behaviour or environment, or by teaching new skills**.
- Programmes are **multicomponent and individualised** to the patient's condition, needs and environment. Possible components are:
 - patient education and self-monitoring with a bladder diary;
 - pelvic floor muscle (PFM) training (PFMT) and exercise, and active PFM contraction to occlude the urethra (stress strategies, "the knack");
 - urge prevention and suppression (urge strategies), and bladder training (BT) with urge control techniques (distraction, self-affirmation);
 - biofeedback (BF) and electrical stimulation (E-stim);
 - fixed or incremental voiding schedules and delayed voiding;
 - fluid management, caffeine reduction, dietary change and weight loss;
 - teaching normal voiding technique and other lifestyle change.
- In most urological and primary care settings, programmes are **designed and delivered by registered nurses and advanced practice providers (APPs)** after a simple non-invasive assessment.
- The **three most established programmes** are **PFMT, behavioural training with urge suppression, and BT**. They rest on the idea that patients can be educated about their condition and can learn skills that reduce or abolish symptoms. They fall into two fundamental approaches.

TWO FUNDAMENTAL APPROACHES TO BEHAVIOURAL TREATMENT

	Outlet-directed	Bladder-directed
Target	Bladder outlet	Bladder function, through voiding habits
Skills taught	PFM strength and control; urge suppression techniques	Changed voiding habits, supported by urge control techniques
Examples	PFMT; behavioural training with urge suppression	Bladder training; delayed voiding

PATIENT-DEPENDENT VERSUS CAREGIVER-DEPENDENT INTERVENTIONS

	Patient-dependent	Caregiver-dependent
Examples	BT, PFMT, behavioural training	Habit training, prompted voiding
Relies on	Active participation of the patient	Caregiver knowledge and motivation
Requires	Adequate function, ability to learn, and motivation	Not dependent on the patient's physical function or mental status

	Patient-dependent	Caregiver-dependent
Suited to	Functionally and cognitively intact patients	Patients with functional and cognitive impairment

- **Lifestyle modifications** act **indirectly**. They change behaviour or remove factors that aggravate symptoms or make the bladder harder to control: fluid management, caffeine reduction, avoiding bladder irritants, managing constipation, smoking cessation and weight loss.
- **Evidence base**. A large body of research supports behavioural treatment. Multiple organisations and international guidelines recommend it for UI and **non-neurogenic overactive bladder (OAB)**.
- The **International Consultation on Incontinence (ICI) Committee on Adult Conservative Management** (7th ICI, 2023) grades every conservative treatment (Tables 114.1 and 114.2) using the **Oxford grading system** (Table 114.3).
- The **Agency for Healthcare Research and Quality (AHRQ)** comparative effectiveness review of non-surgical treatments (2018) is summarised in Table 114.4.
- The chapter also covers **mechanical devices** that prevent urine loss.

Fig. 114.1 — *Common behavioural treatments*. The figure groups the treatments as follows:

- **Core treatments**: patient education; PFMT with stress strategies; behavioural training with urge suppression; BT with urge control strategies; scheduled voiding programmes; lifestyle modifications.
- **Adjuncts**: biofeedback, vaginal cones and E-stim.

Box 114.1 — *Conditions for which behavioural treatments have shown benefit* (modified from Newman and Wein, 2009 and 2013):

- **Storage conditions**: stress UI (SUI); OAB (urgency, frequency, nocturia); urgency UI (UUI); **neurogenic detrusor overactivity**; mixed UI (MUI); nocturia.
- **Pelvic floor and voiding conditions**: **PFM spasm** (dysfunction, high-tone muscle); **non-relaxing striated sphincter**.

TABLE 114.1 INTERNATIONAL CONSULTATION ON INCONTINENCE RECOMMENDATIONS: EDUCATIONAL AND LIFESTYLE MANAGEMENT

Domain	Recommendation	Grade
Prevention	Enhanced education on the pelvic floor for healthy women. It improves knowledge and may reduce symptoms of pelvic floor dysfunction.	C
Weight loss	Behavioural weight loss for overweight and obese women with UI	A
	Lifestyle weight loss for overweight and obese men with UI, particularly those with type 2 diabetes	B
	Weight loss programme for overweight women with UI or prolapse, to improve prolapse symptoms and severity	C
Diet	Adequate fluid intake and caffeine reduction for UI and LUTS	B
	Caffeine reduction in men: intake equivalent to about two cups of coffee a day (250 mg) is associated with UI in men	C
	Diet high in fruit, vegetables and whole grains, with moderate calories, to prevent UI in postmenopausal women	B
	High vegetable, fruit and vitamin C intake in men, for primary and secondary prevention of LUTS, UI and benign prostatic hyperplasia (BPH)	C
	Vitamin D supplementation to reduce UUI episodes in postmenopausal Black women	B
Physical activity	Higher activity (≥43.2 metabolic equivalent [MET] hours per week) to prevent UI in continent, parous, middle-aged women	C

Domain	Recommendation	Grade
	Programme to reduce abdominal fat in young overweight women with UUI, to reduce LUTS. Further high-quality randomised controlled trials (RCTs) are needed.	C
Smoking	No UI or LUTS recommendation possible. Smoking cessation may be offered at the first encounter for general health.	D
	Smoking abstinence for men with UI	C

Modified from Dumoulin et al., ICI 7th edition, 2023.

- **Behavioural weight loss in overweight women is the only grade A lifestyle recommendation.** Every other lifestyle measure is grade B or below.
- **Trap:** smoking cessation is graded D for UI and LUTS in general (no recommendation possible), yet C for men with UI.
- The one caffeine cut-off to know is **about 250 mg a day (two cups) in men.**

TABLE 114.2 INTERNATIONAL CONSULTATION ON INCONTINENCE RECOMMENDATIONS FOR CONSERVATIVE BEHAVIOURAL TREATMENTS (REBUILT BY MODALITY)

Population	Recommendation	Grade
Pelvic floor muscle training in women		
UI (all types)	Offer the most intensive health professional-led PFMT possible within service constraints	A
	Supervised PFMT first-line at all ages	A
	Adding motor learning, abdominal or hip training, intravaginal resistance devices or adherence strategies: no clear benefit, not routine	B
	Adding clinic- or home-based BF: no clear benefit. Consider BF for marked PFM weakness, atrophy or reduced proprioception.	B clinic; B home; A both
	Direct PFMT over indirect methods (Paula method, Sapsford approach, hypopressive training)	B
	PFMT first-line before surgery. Surgery is more effective, but its benefit must be weighed against adverse events and cost.	B
	PFMT and continence pessaries both beneficial; either can be considered	B
	PFMT over pharmacotherapy, because drugs cause more adverse effects and more dropout	B
	Adding vaginal cones (VCs) or E-stim to PFMT adds nothing; not routine	C
Prevention	Supervised PFMT to prevent UI	B
	Continent pregnant women: intensive, supervised PFMT with regular professional contact, to prevent antepartum and postpartum UI	A
	Education to prevent UI in older women	A
Childbearing women	PFMT first-line for UI persisting 2–3 months after delivery	A
	"Intensive" strength training (supervision and content) increases effect	B
	PFMT for all pregnant and postpartum women regardless of continence status, weighing the cost and benefit of population approaches	A antepartum; B postpartum
	Population model. Antenatal: daily home PFMT plus weekly physiotherapist-led classes for 12 weeks from 16–24 weeks' gestation . Postnatal: individually taught strengthening programme with adherence strategies after forceps delivery or a baby ≥4000 g .	C

Population	Recommendation	Grade
SUI	PFMT and VCs both effective. PFMT is favoured because some women cannot use cones and cones have side effects.	B
	Adding VCs to PFMT: no recommendation possible	None
	PFMT better than E-stim first-line; PFMT better than BT first-line	B
	Stress-predominant UI: add PFMT to a continence pessary	B
	PFMT and drugs both effective first-line; PFMT avoids drug adverse effects	B
	SUI with vaginal atrophy: PFMT plus intravaginal oestrogen over oestrogen alone	C
SUI and MUI	Supervised individual or group PFMT, with adequate contraction confirmed, in postmenopausal women	B
	VCs and E-stim equally recommended. Side effects and discomfort limit their utility.	B; D
	Add PFMT after insertion of a midurethral tape	B
SUI and UUI	Add PFMT to drug therapy: tolterodine for UUI, duloxetine for SUI	B
UUI	PFMT and BT first-line	B
	Add PFMT to tolterodine (UUI) but not to duloxetine (SUI)	B
UUI or MUI	Oxybutynin not recommended over first-line PFMT	B
	PFMT and drugs (β -adrenergic agonist, tolterodine) both beneficial. Drugs cause more adverse effects and dropout, so offer PFMT first-line.	B
SUI, UUI or MUI	PFMT over BT alone as first-line	B
	PFMT over E-stim as first-line	A
	Adding PFMT to BT gives additional benefit	C
Neurological disease with OAB	Multiple sclerosis (MS), incomplete spinal cord injury (SCI), Parkinson disease (PD) or after stroke: PFMT alone or combined is effective, safe and first-line	B
Pelvic floor muscle training in men		
Postprostatectomy incontinence (PPI)	Preoperative PFMT instruction gives earlier return of continence after radical prostatectomy (RP)	B
	Early PFMT, preoperative or early postoperative, reduces time to continence	B
	Persistent PPI: adding BF is not routine; consider it for reduced proprioception	B
	Adding E-stim to PFMT is not routine, although combinations beat no treatment	B
	Novel PFM-activating interventions or support groups may help; data very limited	C
	PFMT may reduce climacturia and erectile dysfunction after RP	C
	Pilates is not superior to PFMT, so not recommended over it	B
	Pilates over no treatment may be considered	D
	Pilates, yoga and general exercise have no impact on urinary symptoms after prostate cancer treatment; not routine	D
	Strong PFM contraction immediately after voiding, or urethral massage to empty the urethra	C
Vaginal cones		
General	Assessment by a trained professional is needed, because some women cannot insert or retain a cone and cones cause side effects and discomfort	D
SUI	VCs with supervised training first-line, for those able and prepared to use them	B
SUI or MUI	VCs and E-stim equally recommended; side effects limit utility	B; D
Electrical stimulation		

Population	Recommendation	Grade
SUI	E-stim is better than no treatment or control for symptoms and quality of life (QoL). External and transvaginal E-stim can be used in the short term.	B
	Drug treatments are less effective than E-stim or PFMT plus E-stim; not routine	B
	Adding E-stim to PFMT or to BF-assisted PFMT is not routine	B
	Acupuncture as an optional treatment	C
UII	E-stim can be considered	B
	Propantheline (UII), or oxybutynin and tolterodine (detrusor overactivity [DO]), are less effective than E-stim; not routine	B
SUI and UII	Pudendal E-stim may be more effective than transvaginal E-stim	C
OAB or MUI	Add E-stim to BT, or to BT plus BF	B
UII or OAB	Tibial nerve stimulation (TNS) is better than no active treatment	C
	Transcutaneous and percutaneous TNS are equally effective; transcutaneous has fewer adverse effects	B
	Either form of TNS can be combined with other effective treatments	B
OAB, UII or MUI	Electroacupuncture or anticholinergics	B
Neurological OAB (e.g., stroke)	Transcutaneous electrical nerve stimulation (TENS) as a first-line option	B
Bladder training and toileting assistance		
Women with UI	BT first-line	A
	Most intensive BT supervision possible. Choose training parameters from the operant conditioning literature. BT may be preferred because it has no drug adverse effects.	D
	Timed voiding every 2 hours as sole intervention for mild UI or infrequent voiding	C
	Toileting assistance for OAB and UI after stroke, or in PD or incomplete SCI	C
Women with LUTS	BT for frequency and nocturia	B
	E-stim plus BT over BT alone for nocturia; E-stim (not BF) added to BT for frequency	B
	BT alone and BT plus anticholinergic are equally effective for nocturia with OAB	B
	Moxibustion added to BT for nocturia	C
UII	BT to treat frequency	B
UI or OAB	Insufficient evidence for BT for urgency, or for choosing between BT and drugs for urgency	D
	Either BT or anticholinergics, bearing drug side effects in mind; BT may be preferred	B; D
	Written BT information added to an antimuscarinic does not further improve UI in OAB; not routine	B
	"Simplified" (written) BT added to drug therapy: no extra benefit for urgency, but consider it for frequency	A (one row); B (another row)
With PFMT	BT supplements PFMT in the short term, not the long term (single trial)	C
With E-stim ± BF	Adjunct to BT	B
Conservative management of pelvic organ prolapse (POP)		
Surgery	Liberal, not restricted, postoperative activity instructions	C

Population	Recommendation	Grade
	Adding a physical intervention pre-, peri- or postoperatively does not improve symptoms or severity; not routine	B
Postpartum	Antenatal PFMT not for preventing postnatal prolapse	B
	Postnatal PFMT ± E-stim, or E-stim plus PFMT and BF, to prevent prolapse	C
	Vaginal ring pessary early after delivery: not recommended	B
Symptomatic POP	PFMT reduces prolapse and pelvic floor symptoms ; uncertain whether it reduces stage	A; B
	PFMT plus BF over standard care	C
	PFMT plus BF: no benefit over PFMT alone	D
	Education session with lifestyle and PFMT instruction over verbal instruction alone; postural exercises added to PFMT	D
	Pessary or PFMT, considering cost and adverse events	C
	Pessary plus PFMT over PFMT alone	B
Pessary care	Longer follow-up interval (24 weeks, i.e., 6 months) rather than 12 weeks	C
	Oestrogen plus <i>Lactobacillus acidophilus</i> vaginal tablet: no benefit	C
	Anaesthetic cream over placebo to reduce pain on pessary removal	C

Modified from Dumoulin et al., ICI 7th edition, 2023.

- **What the table is for:** it ranks PFMT against every alternative. **PFMT wins first-line against E-stim (A), BT, VCs and drugs (B). Surgery is more effective, yet PFMT still comes first** because it is less invasive.
- **Pattern across the rows:** adding an adjunct to a correctly supervised PFMT programme (BF, E-stim, VCs, core training) **rarely adds benefit**. BF is reserved for weakness, atrophy or poor proprioception.
- **Traps:**
 - PFMT reduces prolapse **symptoms** (A) but not proven **stage**.
 - Antenatal PFMT does not prevent postnatal **prolapse** (B), although it does prevent **UI** (A).
 - Transcutaneous TNS equals percutaneous TNS in efficacy, with fewer adverse effects.
 - For adding PFMT to drugs, the table gives two answers. One row supports it with both tolterodine and duloxetine. Another supports it with tolterodine only, not duloxetine.

TABLE 114.3 OXFORD LEVELS OF EVIDENCE AND GRADES OF RECOMMENDATION

Grade	Basis
A	Usually consistent level 1 evidence ; often effectively mandatory and built into a care pathway. Excellent evidence may still not earn A if the therapy is prohibitively expensive, dangerous or unethical. A can follow from level 2 evidence, but then needs a larger body of evidence.
B	Consistent level 2 and/or 3 studies, or "majority evidence" from RCTs
C	Level 4 studies, or "majority evidence" from level 2/3 studies, or Delphi-processed expert opinion
D	"No recommendation possible" : evidence inadequate or conflicting, or expert opinion delivered without a formal analytical process such as Delphi

- The system uses **four of the Oxford grades**. A grade can apply **positively (do it)** or **negatively (do not do it)**.
- **Trap:** grade D does not mean "recommended weakly"; it means **no recommendation is possible**.
- **Trap:** Delphi expert opinion reaches grade C, whereas opinion without a formal process is grade D.

TABLE 114.4 RECOMMENDATIONS OF THE AGENCY FOR HEALTHCARE RESEARCH AND QUALITY COMPARATIVE EFFECTIVENESS REVIEW ON BEHAVIOURAL THERAPY

Finding (adult women)	Strength of evidence
Behavioural therapy is more effective than no treatment for cure of SUI	High
Behavioural therapy combined with hormones is more effective than no treatment in SUI	Moderate
For any type of UI, behavioural therapy, anticholinergics and their combination are each more effective than placebo or no treatment	High (behavioural); high (anticholinergic); moderate (combination)
Combination therapy is more effective than no treatment in UUI	Not graded in the source
Behavioural therapy is more likely than anticholinergics to achieve cure in UUI	High

From Balk et al., AHRQ Comparative Effectiveness Review No. 212, 2018.

- **The trap this table sets:** in UUI, behavioural therapy **beats anticholinergics for cure**, with high-strength evidence. Drugs are not the superior first-line treatment for urgency incontinence.

ICI VERSUS AHRQ ON THE SAME FIRST-LINE DECISIONS

Decision	ICI 2023	AHRQ (2008 and 2018 reviews)
Behavioural therapy versus drugs	PFMT over drugs as first-line (B), because drugs cause more adverse effects and dropout	Behavioural therapy superior to anticholinergics for cure in UUI (high strength)
BT versus PFMT	PFMT preferred over BT alone (B)	No difference between BT and PFMT, or between BT and BT plus PFMT (2008 review)
Behavioural therapy versus no treatment	PFMT and BT first-line (A)	Superior for SUI cure (high strength)

Concept — *Conservative therapy is first-line not because it is the most effective option (surgery is more effective for SUI) but because it is risk free and at least as effective as drugs. Every recommendation in these tables follows from that asymmetry of harm.*

INDICATIONS

- Behavioural interventions are well established for **SUI, UUI, MUI and OAB**. They are also appropriate for **voiding dysfunction**, although the evidence there is weaker.
- Most patients improve to some degree, but outcomes vary widely and **responders cannot be predicted**. Because the treatment is virtually risk free, it should be **offered first-line to anyone willing and able to participate**.
- Even patients with **dementia** can benefit, from an appropriate **caregiver-guided** programme.

Assessment Before Behavioural Treatments

- The most important step is a **focused, detailed history**: onset, duration, progression, characteristics and **situational triggers**. These determine how the programme is designed and implemented.
- A **3- to 7-day bladder diary or frequency-volume chart** supplements the history (Fig. 114.2 shows a completed example).
 - It records the type and amount of fluid intake, the type and frequency of symptoms (UI episodes), voiding frequency and volume, associated urgency, and the circumstances of each leakage.
 - Pad type and number can be added to quantify leakage.

- The diary is the **best non-invasive tool for objectively monitoring voiding habits and treatment effect**, and it guides which treatment components to use.
- Voided volumes are more burdensome to record but give a practical estimate of **functional bladder capacity** in daily life.
- Completing the diary has a **self-monitoring effect**. The patient becomes aware of drinking and voiding habits and sees how leakage relates to activities. This is thought to empower retraining, and it shows the times and activities of highest risk, so that skills can be used pre-emptively.
- The **Questionnaire-Based Voiding Diary (QVD)** is a validated alternative. It self-assesses fluid type and volume and UI type, and is completed at the first office visit in **5–7 minutes** (Fig. 114.3).

Fig. 114.3 — *Questionnaire-Based Voiding Diary* (Arya et al., 2008). The recall period is the **last month**. It has four sections and a clinician-rated severity index.

- **A, beverage intake:** ten beverage types (water; caffeinated and decaffeinated coffee, tea and soda; milk; juice; alcohol), each as drinks per day (0 to 10+) and size (under 8 oz to over 24 oz).
- **B, intake behaviour** (five-point scale, never to all of the time): large caffeinated or carbonated intake, extra fluid for weight, drinking when not thirsty, **restricting fluid to control symptoms**.
- **C, output:** large voids on waking, afternoon, evening and night.
- **D, symptoms:** daytime frequency bands (1–5 to over 20); nocturia (0 to 4+); urgency; urgency leakage; stress leakage; leakage frequency and amount.
- **Severity index** (slight to very severe), completed by the clinician.

Concept — *Assessment before behavioural therapy is aimed at triggers and habits rather than diagnosis. The diary is simultaneously the assessment tool, the outcome measure and a treatment in its own right through self-monitoring.*

PATIENT EDUCATION

- Behavioural treatment rests on **principles of learning**, so the first step is basic education about the condition, the treatment process and its goals.
- Education covers the **anatomy and function of the bladder, urethra, vagina or prostate, PFMs and rectum**, how they interrelate, and the causes and mechanisms of the patient's own condition.
- The patient must understand that the programme means **changing habits and learning skills**, and that **improvement is gradual**. Knowing that results depend on active participation and daily practice improves adherence and sets realistic expectations.

Pelvic Floor Muscle Training

- PFMT and exercise, also called **pelvic floor rehabilitation**, is a **cornerstone of behavioural treatment** for UI, urgency and frequency.
- **History.**
 - **Margaret Morris (1936)** first described it, teaching women to control and exercise the perivaginal muscles to strengthen them and reduce SUI.
 - **Arnold Kegel (1948)**, a gynaecologist, popularised PFM exercises (PFME). He proposed that SUI arose from **lack of awareness of PFM function and coordination**, and showed that repetitive contractions to build strength reduced SUI.
 - PFMT has since evolved into a combined **nurse-directed behavioural treatment and physical therapy**.

- **Proposed mechanism in women** (causal chain):
 - a contraction **raises the urethra and presses it against the symphysis pubis**, preventing urethral descent and supporting the pelvic organs;
 - training causes **hypertrophy of the striated muscle**, increasing external mechanical pressure on the urethra;
 - intensive PFMT is thought to **reinforce bladder neck support**, limiting its descent when abdominal pressure rises.
- PFMT has **three elements**: learning to contract and relax the PFMs; a regular exercise regimen for strength and control; and **actively contracting to occlude the urethra** during activities that raise abdominal pressure. The details of regimens vary.

Assessment of Pelvic Floor Muscle Function

- Baseline assessment of strength, coordination and control uses **visual observation** (at rest and with Valsalva), **transvaginal digital palpation**, electromyography (EMG), manometry or ultrasound.
- **Visual contrast in women**:
 - **correct contraction = perineal elevation**: inward (ventrocephalad) movement of the vulva, perineum and anus;
 - **incorrect contraction = perineal descent**: excessive dorsocaudal movement.
- **Digital assessment at rest and during contraction is the technique most used in practice.**
- Manual strength scales:
 - the **Oxford Grading System** is the most commonly used, and adapted versions have been studied;
 - the **Brink score** is a **four-point scale of contraction pressure, vertical displacement and endurance** of the squeeze.
- **Digital technique** (Fig. 114.4):
 1. Insert one or two fingers into the vagina in the mid-sagittal plane for **6–8 cm**.
 2. Palpate the levator ani just above the hymeneal ring, at the **midline and at 4 and 8 o'clock**, for strength and for tenderness.
 3. Assess at rest, during contraction and during relaxation: ability to relax, duration and symmetry of contraction, muscle bulk, resting tone and strength.
 4. Ask for a maximal contraction held as long as possible, then short fast contractions ("**flicks**").
 5. Use tactile feedback (tapping, a slight quick stretch) to help identification when the patient cannot find the muscle, does not understand how to contract it, or has very weak PFMs.
- **Trap**: when asked to contract, women may use the **wrong muscles, strain down (Valsalva)**, or fail to activate all layers. The examiner watches for accessory contraction of the **gluteal or abdominal muscles**.
- **Brink's three criteria**, each recorded:
 - **pressure**, from imperceptible to a firm squeeze;
 - **duration** in seconds;
 - **alteration in position**: a well-supported pelvic floor lifts the base of the examiner's finger.
- Note any **co-contraction** (abdominal movement, gluteal lifting) and the degree of any **PFM overactivity** (voluntary relaxation absent or partial).

	Overactive PFM (hypertonicity)	Hypotonicity
Tone	Increased muscle tone from contractile or viscoelastic components	Decreased muscle tone from contractile or viscoelastic components

	Overactive PFM (hypertonicity)	Hypotonicity
Components	Elevated contractile activity and/or passive stiffness	Reduced contractile activity and/or reduced passive stiffness
Functional sign	May contract when relaxation is needed , for example during micturition	Weak contraction

- **Anal assessment**, an alternative route to the PFM:
 1. The patient relaxes and bears down.
 2. As the sphincter relaxes, gently insert an index finger pointing towards the umbilicus. A normal sphincter closes snugly around the whole finger.
 3. The **puborectalis** is palpable about **2.5–4 cm from the anal verge**.
 4. The patient tightens the rectum; a gripping or "pulling in" should be felt around the entire circumference.
 5. Resting tone is recorded as weak, moderate or strong.

Fig. 114.4 — *Pelvic floor muscle assessment*. The figure uses a clock face: the pubic bone is at 12 o'clock (anterior), the anus at 6 o'clock (posterior), and the **levator ani** at **4 and 8 o'clock**.

Teaching Pelvic Floor Muscle Control

- The first step is for the patient to **identify the PFMs and contract and relax them reliably**. This is taught with verbal feedback during vaginal or anal palpation, with visual or auditory BF, or with E-stim.
- **Typical instructions:**
 - "Contract around my fingers; pull up and in."
 - "Without tensing legs, buttocks or stomach, tighten the ring around the anus as if stopping gas or a stool; feel closing and lifting."
 - Men: "Move the penis up and down without moving anything else."
 - Women: "Feel the vagina and rectum pull up and in."
- The commonest problem is **muscle isolation**, and many women lack confidence that they are contracting correctly.
 - Patients recruit the rectus abdominis or gluteals instead of, or as well as, the PFMs (**co-contraction**).
 - Some women substitute a **straining Valsalva**, which **threatens to worsen the very condition being treated**.
 - Some abdominal contraction is counterproductive because it raises pressure on the bladder and pelvic floor.
- The clinician therefore teaches **selective PFM contraction with ancillary muscles relaxed**. Telling the patient not to hold her breath, or to count aloud, helps avoid bearing down.
- Some clinicians add **transversus abdominis** training, believing it facilitates PFM contraction. Evidence on whether abdominal or hip training adds efficacy is **conflicting**, and a scoping review found little evidence for the proposed core-strengthening mechanism.
- **Verify correct contraction before starting the exercise regimen**. Inability to identify or exercise the PFMs correctly may be the **commonest reason for poor outcome**. Control is a skill that takes time, but most patients succeed with repeated training.
- The old approach was a pamphlet or brief instruction to "lift the pelvic floor" or interrupt the stream. It suffices for some patients but **does not ensure the correct muscles are used** at home.

Concept — *The usual point of failure in PFMT is muscle identification, not exercise dose. Every teaching step exists to separate a true PFM lift from co-contraction and Valsalva.*

Pelvic Floor Muscle Exercise Regimens

- Once correct contraction and relaxation are shown, daily exercise has **two purposes: strength and motor skill**.
- The **ideal regimen is not established**. The ICI could make no robust recommendation on generic versus individualised exercise, submaximal versus near-maximal contraction, daily versus three times weekly, or adding upright positions. Good results have come from **45–60 paired contractions and relaxations per day**.
- The authors use an "**exercise prescription**" (Box 114.2):
 - **quick flicks of 1–2 seconds**, then **endurance contractions of 5 seconds building to 10 seconds**;
 - **high concentrated effort** on each contraction, because greater intensity is associated with greater strength gain.
- **Relax completely between contractions, with a work-to-rest ratio of 1:1 or 1:2**, to allow recovery and optimise strength building.
- Spread the exercises over **2–3 sessions a day** to avoid fatigue.
- **Start reclining**, the least challenging position, then **progress to sitting and standing** so that the skill works in any position.

Box 114.2 — *Sample pelvic floor muscle exercise prescription* (modified from Newman and Wein, 2009 and 2013):

- **Session:** lying → sitting → standing. In each position do **5 quick contractions of 1–2 seconds** and **5 sustained contractions of 5 seconds building to 10**. Rest after each for as long as the contraction lasted. The three positions make one session of 30 contractions.
- **Dose:** **two sessions a day**, morning and evening, **60 exercises in total**.
- **Tips:**
 - empty the bladder before starting;
 - **count aloud** during long holds so the breath is not held;
 - keep the abdomen, legs and buttocks relaxed — a hand resting on the abdomen should not move;
 - breathe through each contraction, and take a deep breath between exercises if it helps.

Use of Pelvic Floor Muscle Contraction to Prevent Stress Urinary Incontinence

- The goal is to prevent leakage in daily life by **volitionally occluding the urethra with a PFM contraction**. Exercise alone improves urethral pressure and support; this additional motor skill lets the patient close the urethra consciously when loss is imminent.
- The history or diary identifies the circumstances of leakage. The patient then learns to anticipate them and **contract before and during** coughing, sneezing, lifting or any other precipitating activity.
- This **anticipatory contraction** is called the **stress strategy** or "**the knack**". An anticipatory contraction reduces the volume of cough-related leakage immediately.
- It needs vigilance and conscious effort at first but **eventually becomes automatic**.

Box 114.3 — *Handout for teaching stress strategies and "the knack"* (modified from Burgio et al., 1989; Miller et al., 2020; Newman and Wein, 2009):

- **Anatomy and cause, for the patient:** the PFMs span pubis to coccyx and ischial tuberosity to tuberosity, closing the bladder and bowel outlets (female and male diagrams). Leakage occurs when outward pressure exceeds closing pressure (cough, sneeze, standing, bending, lifting) or during urgency on the way to the toilet.

- **Steps:**

1. Squeeze quickly **just before and during** the triggering activity.
2. If you forget and leak, squeeze anyway. This will not stop that leak, but it links the squeeze to the trigger.
3. Timing takes practice; with time it becomes automatic.
4. **"Squeeze before you sneeze"; "squeeze throughout coughing."**

- Once habitual, the same squeeze can be used to **delay voiding and prevent leakage after urgency**. A bladder diary identifies the triggers.

Evidence for Pelvic Floor Muscle Training

- A very large body of evidence supports PFMT.
 - Cochrane review: **effective for SUI and MUI** and can reduce urgency; women with **pure SUI may do best**.
 - **PFMT beats no treatment, placebo drug or inactive control for SUI, UUI and MUI**, on cure, QoL, daily leakage episodes and pad test leakage, both immediately and long term.
- The chapter's pooled figures are **cure 16–27%** and **improvement 48–80.7%**.
- The best approach is unknown, but there is consensus that **supervised PFMT is more effective than unsupervised**. The degree and type of supervision needed are uncertain.
- The effect is enhanced by sound training principles: **specificity, overload progression, confirmation of correct contraction before training, and active use of the PFM to occlude the urethra** (the knack).
- **Age is not a deterrent**. The effect in SUI does not diminish with age; outcomes in older women match those in younger women. Therefore **supervised PFMT is first-line for women of all ages with SUI, UUI or MUI**.
- **Pregnancy and postpartum:**
 - A 2020 Cochrane review found that a structured PFMT programme offered to **continent antenatal women early in pregnancy** may prevent UI in late pregnancy and postpartum.
 - Postpartum women with UI randomised to professionally taught, supervised PFMT were **less likely to be incontinent 6–12 months after delivery** than those given no treatment or usual care.
 - **ICI: PFMT first-line for postpartum UI persisting after delivery**. The text gives this as 3 months; Table 114.2 gives 2–3 months. The source does not explain the difference.
- **Men after RP:**
 - SUI complicates RP **whatever the technique**. **PFMT is the primary conservative treatment for PPI**, taught by digital rectal examination (DRE) with verbal feedback, or with BF or E-stim.
 - Trials in established PPI are mixed but generally show that PFMT **reduces UI in the first 3 months**; persistent long-term UI is less studied.
 - In an RCT of men with UI persisting 1–17 years after RP, **8 weeks of behavioural training** (PFMT plus bladder control strategies) reduced UI episodes significantly more than control, and the benefit was sustained to 12 months.
 - Adding E-stim or BF to PFMT **does not improve continence**. However, BF is an effective teaching approach, and providers may prefer it to teaching by DRE.
- Perioperative PFMT to prevent PPI is covered under Prevention of Urinary Incontinence.

Concept — PFMT works by two routes: **structural** (hypertrophy and bladder-neck support built by exercise) and **skill** (timed urethral occlusion, the knack). **Supervision and confirmed correct contraction** are what

separate effective programmes from ineffective ones, which is why supervised PFMT is grade A and add-on gadgets are not.

Vaginal Cones for Pelvic Floor Muscle Training

- Weighted vaginal cones were developed to **test PFM function**. They are used to help patients identify and control the PFMs and to support adherence (Fig. 114.5). They provide **progressive muscular overload**.
- **Mechanism:** with the cone in the vagina and the patient standing, the **sensation of the cone slipping** gives strong sensory feedback that prompts a PFM contraction to retain it.
- **Protocol:**
 1. Insert the heaviest cone that can be retained standing.
 2. Then retain it while walking.
 3. Increase cone weight and retention time step by step.
 4. Goal: retain the cone while walking for **2 sessions of 20 minutes a day**. Rising weight maintains overload.
- **Limitations:** discomfort; difficult insertion with a **narrow introitus**; retention impossible with **POP or a wide introitus**.
- **Trap:** the vagina is **not vertical**, so some women retain the cone **without contracting the PFMs**. Its value as a PFMT aid is therefore questionable.
- **Evidence.** A Cochrane review of small trials of uncertain quality found some evidence that cones are **better than no active treatment for SUI**, and possibly similar to PFMT and PFM E-stim. More definitive trials are needed. An overview of Cochrane reviews found moderate- or high-certainty evidence that cones may beat no treatment or placebo.

Other Pelvic Floor Muscle Training Devices

- **Colpexin sphere:** a weighted device for women with **POP and UI**.
 - It is a smooth medical-grade polycarbonate sphere with a braided string. It is self-inserted, by applicator or by hand, into the vaginal vault **above the levators** (Fig. 114.6).
 - By **elevating the prolapse**, it makes a proper PFM contraction easier.
 - It is fitted by size (**32, 36, 39, 42 and 44 mm**), degree of POP and PFM tone.
 - Exercises are done at least twice daily with the sphere in place. To progress, the patient tugs gently on the string while contracting.
 - In a single-blind RCT in stage I–II POP, sphere plus exercise versus exercise alone showed **no difference in PFM strength at 16 weeks**. Vaginal symptoms improved in both groups and **POP stage did not change**. Adverse events were mainly vaginal discomfort, with discharge and abdominal discomfort.
- **Motivation** determines outcome, because the patient must exercise correctly and consistently. Perceived competence in an activity motivates intrinsically, and **mastering a goal** is most strongly linked to self-perception, autonomous motivation, enjoyment and persistence. BF built into training devices, and pelvic floor trainers, help patients feel confident that they are exercising correctly and motivate them to continue.
- **Home PFMT devices** are now numerous. Many claim to guide exercise, and some give feedback on contractions and counts. Many, however, **lack progressive resistance and accurate feedback with standardised instruction**. A practical, safe device with these features is still needed.

Behavioural Training With Urge Suppression

- A **stand-alone treatment** that teaches patients to **use the PFMs to inhibit detrusor contraction**, within a larger urge suppression strategy, as a new way of responding to urge. It is used for urgency, frequency and UUI.
- **Origin** (sequence of discovery):
 1. BF-assisted training used bladder and anal sphincter BF together during retrograde filling, to teach cortical inhibition of detrusor contraction.
 2. It was noticed that **spontaneous anal sphincter activity altered detrusor pressure**.
 3. A well-timed voluntary anal sphincter (PFM) contraction under visual BF was then shown to **abort fully developed detrusor contractions, deter developing ones and suppress urgency**.
 4. Later, the skill was shown to be teachable **without BF**, using verbal feedback from vaginal palpation. In an RCT in older women with UUI this gave outcomes similar to bladder–sphincter BF when built into a larger strategy.
- Consequently **PFMT, designed for SUI, is now central to the treatment of UUI and OAB**. PFM control is taught exactly as for SUI; what differs is **how the muscle is used** — as part of a broader response to urge, not only to occlude the urethra. Urge suppression with incremental interval increases is also integral to BT.
- **The "gotta go" response is counterproductive**. Rushing raises physical pressure on the bladder, heightens fullness, worsens urgency and triggers detrusor contraction. Approaching the toilet adds **visual cues that trigger urgency and UUI**, a conditioned response.
- **Urge suppression strategy** (Fig. 114.7):
 1. Do not rush; stay away from the bathroom and its cues.
 2. Pause, sit down if possible, and relax the whole body.
 3. **Contract the PFM repeatedly and quickly, without relaxing in between.**
 4. Concentrate on inhibiting the urge and let it pass, using distraction if it persists.
 5. When it subsides, **walk at a normal pace** to the toilet.
- **After mastery**, delay voiding by **5 minutes** to consolidate the skill. In frequent voiders, lengthen the delay gradually until voiding frequency is normal.
- **Efficacy**. As stand-alone therapy for UUI, it is established in clinical series and intention-to-treat RCTs, with mean reductions in incontinence of **60–80%**.
- It also reduces **urgency, frequency and nocturia in both men and women**.
- **Nocturia protocol**: on waking with fullness or urge, use urge suppression. If the urge subsides, go back to sleep. If it has not subsided after a minute or two, get up and void, so as not to disturb sleep unnecessarily.
- **Combination with drugs** improves outcomes of drug therapy for UUI (see the trial table).
- The urge suppression strategy is also used in **BT and delayed voiding**.

BEHAVIOURAL TRAINING VERSUS AND WITH DRUG THERAPY — TRIAL SERIES

Trial	Population	Comparator	Verdict
First RCT of behavioural training in women	Older women with UUI	Individually titrated drug therapy	Behavioural better : fewer UI episodes, more perceived improvement and satisfaction

Trial	Population	Comparator	Verdict
MOTIVE (Male Overactive Bladder Treatment in Veterans) — first RCT in men	Men with OAB without bladder outlet obstruction	Extended-release oxybutynin	Equivalent significant reduction in 24-hour voids
Men, frequency	Men with OAB	Antimuscarinic plus α -blocker	Lower 24-hour frequency with behavioural training
Nocturia, women	Women	Drug therapy; placebo	Both beat placebo; behavioural better than drug
Nocturia, men on α -blocker	Men with OAB	Added antimuscarinic	Adding behavioural better than adding antimuscarinic
Nocturia, men	Men with nocturia	α -blocker	Behavioural alone or with α -blocker gave large reductions
National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) Urinary Incontinence Treatment Network	Women with urgency-predominant UI	Tolterodine extended release alone versus with behavioural training	Combined better on perceived improvement, satisfaction, distress and bother, on drug and 6 months after stopping it
Men, frequency (combination trial)	Men with frequency	Drug alone; behavioural alone	Combined better than drug alone, not than behavioural alone; best on secondary outcomes

Fig. 114.7 — Patient handout for urge suppression and bladder training.

- **Urge wave:** grows, peaks, subsides, stops; the response is the strategy above, with about **5 quick squeezes**, then a 5-minute wait before walking slowly to the toilet.
- **Voiding schedule:** postpone voiding by 5 minutes → then by 15–20 minutes → until able to wait **3–4 hours**.

Concept — Urgency is a wave that peaks and passes, and a PFM contraction reflexly inhibits the detrusor. Behavioural training replaces the conditioned rush with stillness and a squeeze. Every urge-directed technique in the chapter — BT, delayed voiding, nocturia management — reuses this one manoeuvre, which is why behavioural training matches or beats antimuscarinics.

THE ROLE OF BIOFEEDBACK

- **BF is not a treatment in itself but a technique for teaching control of a physiological response.** It gives precise, instantaneous feedback, for example of a PFM contraction. It is also used for headache, muscle spasm and hypertension.
- **Kegel's perineometer** (Fig. 114.8) was a **vaginal pressure BF device** — a manometer gauge connected to an intravaginal pressure probe — that gave women immediate feedback on their attempts to contract.
- BF is especially useful for three groups: patients who **cannot identify or isolate the correct muscles**; tracking strength during treatment; and patients who need encouragement to continue.
- **Modern systems** are computerised with visual display (Fig. 114.9).
 - PFM activity is measured by **manometry or surface EMG (sEMG)**, using vaginal or anal probes or surface electrodes.
 - The signal is amplified and shown as a graph (Fig. 114.10) or played as sound.
 - Learning occurs by **operant conditioning (trial and error)** as the patient watches the result of each attempt (Figs. 114.11 and 114.12: a woman reclining and a man sitting while viewing the monitor).

	Manometric pressure BF	EMG BF
Signal	Intravaginal or anal pressure	Muscle electrical activity
Positioning	Probe-dependent	Lightweight electrodes that stay in place (with adequate filtering) allow more functional positions
Channels	Single	Multichannel: PFM contraction shown alongside inhibition of accessory abdominal and gluteal muscles

- **Evidence.**

- BF-assisted PFMT within multicomponent conservative programmes produces significant improvement in UI.
- Whether BF **adds** to PFMT is **inconsistent**. A Cochrane review found PFMT plus BF more likely to give cure or improvement, but the **groups differed in clinician experience and in contact time**, since BF groups had more provider contact.
- A large multisite RCT in SUI or MUI found **no difference between PFMT plus EMG BF and PFMT alone**. Recent systematic reviews agree that there is **no clear benefit of adding BF**.
- BF nevertheless remains an **excellent method for identifying and exercising the PFMs**, particularly for weak PFMs, inability to identify the muscle (reduced proprioception), or uncertainty about correct contraction. It is widely used in urology, urogynaecology and PFM physiotherapy.

Fig. 114.10 — Biofeedback graph (EMG in microvolts). It contrasts two contraction types:

- **fast-twitch fibres** — quick contractions of about 2 seconds, seen as sharp spikes;
- **slow-twitch fibres** — sustained contractions of about 10 seconds, seen as plateaus;
- in both, the trace returns to baseline between contractions.

Concept — BF improves **learning, not outcome**. Its value lies with the patient who cannot find the muscle. Once contraction is correct, BF adds no efficacy, and its apparent benefit in older trials reflected extra clinician contact.

Home-Biofeedback Devices

- Intravaginal systems for home use give **real-time verification** that the exercises are performed correctly and consistently, by guiding PFM elevation with visual feedback.

	PeriCoach	Leva (Axena Health)
Sensor	Silicone-coated vaginal perineometer (pressure)	Vaginal insert with accelerometers (motion) ; does not measure pressure
Key property	Pressure rises with both a correct contraction and a Valsalva	Distinguishes a correct upward and inward lift from Valsalva , which deflects the device downward
Support	App prompting two sets a day	Bluetooth smartphone app with reminders, plus scheduled calls or texts from a trained coach

	PeriCoach	Leva (Axena Health)
Evidence	RCT against supervised pelvic floor physical therapy (four in-person sessions): non-inferior at 3 months on the International Consultation on Incontinence Questionnaire-Short Form (ICIQ-SF)	RCT of PFMT plus Leva versus PFMT alone (SUI or stress-predominant MUI): stopped early for since-corrected technical problems, but fewer SUI episodes with Leva. Fully remote ("virtual") RCT against a home programme with written and video instruction: better Urogenital Distress Inventory (UDI) scores and fewer SUI episodes , more patients "much or very much improved" on the Patient Global Impression of Improvement (PGI-I), no device-related serious adverse events; 8-week intervention with superiority sustained at 6 and 12 months.

Concept — *The discriminating property is what the device senses. A pressure device cannot tell a lift from a strain; a motion device can. That matters because Valsalva substitution is the commonest error in PFMT.*

Pelvic Floor Muscle Electrical Stimulation

- **Definition:** PFM E-stim with a **non-implanted** device applies low-grade electrical stimulation to the PFM. It is used for various UI types and for urgency, frequency and nocturia.
- **Twofold action: PFM contraction and inhibition of unwanted detrusor contraction.**
- It is delivered by clinic machines or portable battery units, in the office or at home. Units vary in current type, waveform, frequency, intensity, electrode type and placement.
- It is usually one component of PFM rehabilitation, where it **helps PFM identification**. It can build strength in LUTS due to **neurological disease (e.g., MS)**. It is often given after failure of other behavioural and medical treatment, with home units for long-term use.

THE TWO MAIN TYPES OF ELECTRICAL STIMULATION

	Long-term (chronic) E-stim	Maximal E-stim
Intensity	Below the sensory threshold	High intensity, just below the pain threshold
Aim	Detrusor inhibition by afferent pudendal nerve stimulation	Urethral closure by direct and reflex contraction of striated periurethral muscle (detrusor inhibition also suggested)
Mechanism	Reflex activation of hypogastric efferents and central inhibition of pelvic efferents, which are sensitive to low-frequency stimulation	Direct and reflexogenic muscle contraction
Dose	6–12 hours a day for several months	15–30 minutes , several times a week or once or twice daily at home

- **Delivery:** in conservative therapy, E-stim uses surface electrodes — transcutaneous stimulation through **suprapubic, sacral or perianal skin electrodes** — or **intravaginal or intrarectal probes**.
- **Parameters** are varied by UI type and stimulation type: current source, pulse width and duration, intensity, frequency, pulse shape, session length, number of sessions, and rest-to-work ratio.

STIMULATION FREQUENCY BY INDICATION

Indication	Frequency
Urgency UI	5–20 Hz (low)
SUI, including PPI	20–50 Hz (higher)
MUI	About 20 Hz, or high and low frequency alternately

Indication	Frequency
All indications	Pulse duration 200–1000 µs ; rectangular pulse; biphasic preferred

- **Evidence in women:**

- Effective for SUI, and for urgency and MUI against sham or no treatment; effective compared with PFMT or BT.
- It **does not enhance outcomes over PFMT alone or PFMT with BF**.
- Its niche is women **who cannot contract the PFMs voluntarily**: 8 weeks of intravaginal E-stim with concurrent voluntary effort improved the ability to contract and reduced UI severity and impact.

- **Men:** less studied. There is little evidence that E-stim improves PFMT with or without BF, but it **may hasten recovery of continence after prostatectomy**.

Concept — *Stimulation is dosed by goal. Sub-sensory, low-frequency, prolonged stimulation targets detrusor inhibition through pudendal afferents; near-painful, higher-frequency, brief stimulation targets sphincter contraction. Because volitional PFMT already achieves both in a patient who can contract, E-stim earns its place only when volitional contraction is absent.*

BLADDER TRAINING AND SCHEDULED VOIDING REGIMENS

- Scheduled voiding regimens, or **toileting programmes**, have for decades been a mainstay of treatment for UUI and OAB. They comprise **BT, timed voiding (TV), habit training (HT) and prompted voiding (PV)** (Table 114.5).
- All share a toileting schedule. They differ in:
 - the **patient's role (active versus passive)**;
 - how the schedule is adjusted;
 - the nature of patient education and the use of reinforcement;
 - the nature of the patient–provider interaction.
- **BT is self-administered**: the patient must resist urgency and delay voiding, and it includes **cognitive and psychological components**. TV, HT and PV mostly **rely on a caregiver**.

Bladder Training

- BT is a broad term covering bladder retraining, bladder discipline, bladder re-education and **bladder drill**. It was developed for **UUI**.
- **The cycle BT breaks**:
 - urgency leads to frequent voiding, which relieves the urge immediately;
 - frequent voiding becomes a habit;
 - the habit leads to **reduced functional bladder capacity**, DO and sometimes UUI;
 - DO produces more urgency, perpetuating the cycle.

It is broken by BT or by progressive delayed voiding.

- **Goal**: restore normal bladder function and capacity through education plus a mandatory or self-adjustable schedule that **progressively lengthens the interval between voids**.
- Originally **bladder drill** was intensive and often inpatient; it is now implemented gradually in outpatient, rehabilitation and other settings.
- BT has three components: **education on lower urinary tract (LUT) function, incremental voiding schedules, and urge control techniques** to postpone voiding and keep to the schedule.

	Standardised schedule	Tailored schedule
Basis	Fixed and predetermined	The patient's average voiding interval from the diary
Start	Half-hourly or hourly	Individual
Progression	Predetermined steps of 5, 10, 15 or 30 minutes	Similar progression

- **BT protocol:**

1. The patient completes a bladder diary.
2. Set an initial interval, often **1 hour while awake** (30 minutes or less at first if needed), or the longest interval that is comfortable.
3. Void on waking, every time the interval passes, and just before bed.
4. **Increase by 15–30 minutes a week**, according to tolerance: fewer UI episodes than the previous week, minimal schedule interruptions, and control over urgency.
5. Target: an interval of **3–4 hours** (Fig. 114.7).

- **Coping with urgency while waiting:**

- relaxation, or **distraction** with an activity needing mental but not physical effort — a game on a phone or tablet, reading, a to-do list. This reduces anxiety and gives the urge time to subside;
- **self-affirmations:** "I am in control of my bladder", "I can wait".

- **Urge suppression within BT:** a few quick PFM contractions and/or a sustained contraction.

- PFM contraction produces **immediate detrusor relaxation**, shown in patients with UI and in patients with and without OAB.
- Proposed mechanism: the quick contraction **compresses the urethra mechanically for a few seconds**, inhibiting the detrusor contraction and reducing the desire to void.
- Hence **repeated quick flicks, with the patient posturally still, waiting for the sensation to pass**, are a useful adjunct to BT.

- Ideally the clinician **reviews weekly**, adjusts the interval and gives positive reinforcement. The patient's voiding diary tracks adherence and progress and shows when to change the schedule.

- **Mechanism is unknown.** The central feature is that BT **dissociates voiding from urgency**: voiding by the clock rather than on urge may weaken the urge–void response. Other hypotheses are:

- improved cortical control of bladder activity and urethral closure;
- improved central modulation of afferent input;
- behaviour change from better awareness of LUT function;
- increased "reserve capacity" as patients learn what provokes leakage.

Evidence for Bladder Training

- The first RCT of BT was in **older women with UI**. Several systematic reviews have since graded the evidence.

- **ICI** (trials of UI of all types and of OAB without UI, mostly in women, few in men):

- BT protocols vary remarkably, and there is **no evidence for a best method or parameter set**. The ICI recommends the **most intensive BT supervision possible**.
- BT is **more effective than no treatment** in UUI, SUI and MUI.
- There is **insufficient evidence in men**.
- BT may work in **motivated frail older women who are cognitively and physically intact**.

- Evidence is insufficient to compare BT with current drugs, or to judge their combination, in DO or UUI.
- A multicentre RCT in **continent, cognitively intact older adults hospitalised with a fall-related hip fracture** gave BT with prompting in hospital plus a telephone reinforcement call 10 days after discharge. It **reduced the incidence of functional UI** at 6 months.
- The **AHRQ** review found **no difference between BT and PFMT**, nor between BT alone and BT plus PFMT.

Concept — *BT works by uncoupling the void from the urge. The schedule is the vehicle and urge suppression is the engine, which is why the protocol advances the interval only as fast as urgency control allows.*

Other Scheduled Voiding Regimens

- Toileting programmes are the **cornerstone of continence care for cognitively impaired patients**. Apart from BT they are **caregiver dependent**: a professional or family caregiver must assist with toileting.
- They benefit toileting-dependent nursing home residents, and people at home who have an available, willing caregiver.
- **Toileting disability** means difficulty with toileting, or needing human or mechanical help with it. Such patients may have mobility or cognitive impairment and need help from at least one person, but **can cooperate** with toileting.
- The choice among TV, HT and PV depends on **cognitive and functional status, variability of the voiding pattern**, and the **need for reinforcement** to maintain adherence.

Timed voiding (also called scheduled, routine or fixed toileting)

- **Goal**: prevent incontinence by giving regular opportunities to empty the bladder **before it reaches capacity**. This reduces leakage from full-bladder detrusor contraction and from SUI; no other physiological mechanism has been identified.
- **Use**: patients who cannot toilet independently. It is used mainly in institutions as a **passive** programme — the caregiver takes the patient to void **every 2–4 hours, except at night** — and in **neurogenic bladder after SCI**.
- The schedule should ideally rest on an objective measure: a bladder diary, a bladder volume instrument, or an electronic monitor of incontinence episodes.
- It is passive because it runs **whether or not the patient feels the need to void**, usually during waking hours only. The goal is dryness; **no attempt is made to teach resisting the urge**.
- **Outcomes**: about **20% of frail elders become completely dry** and **30–50% improve**, with fewer and smaller episodes; home care is probably similar.
- TV also suits **cognitively intact elders with impaired bladder sensation** from neurological disease: voiding by the clock, before urgency or spontaneous emptying, prevents or reduces UI.
- A **2- to 3-day trial of TV** shows whether it will succeed. TV also normalises frequency in **infrequent voiders or those with diminished sensation**.

Habit training

- HT is a **variant of TV** in which the schedule is **matched to the patient's own voiding pattern** rather than fixed.
- From the diary, the interval is set **shorter than the patient's usual interval** and timed to **precede expected incontinent episodes**. The interval can therefore lengthen or shorten through the day, pre-empting UI.
- Traditionally it is **caregiver dependent**, started and maintained by a professional or family caregiver.
- It is used mainly in institutions with cognitively and physically impaired adults, and has been tested in the homebound elderly.

- It also suits **unimpaired adults whose UI recurs at about the same time each day** or is **diuretic induced**. The interval can be fixed permanently or lengthened gradually.
- **Pattern urge-response toileting (PURT)**: 3 weeks of electronic voiding logs → individual schedules → caregivers (mainly spouses) taught when and how to toilet homebound frail elders → **fewer incontinent episodes and less leaked volume**.

Prompted voiding

- **Definition**: PV combines scheduled voiding with **caregiver prompting and contingent social approval**. It teaches people, with or without cognitive impairment, to **initiate their own toileting by asking for help**, and reinforces them when they do.
- **Setting**: mainly nursing homes with cognitively and physically impaired older adults; also home care.
- **Three components**:
 1. **Monitoring**: regular checks, asking the person to report whether wet or dry.
 2. **Prompting**: asking whether they need to void, and assisting, on schedule.
 3. **Praise**: positive feedback when continent and when they try to toilet.
- **Suitable** for older adults with **UII, MUI or functional UI**, whether cognitively intact or impaired.
- **Requirements**: a **consistently available caregiver** is the main one. The individual must also be able to **delay voiding and cooperate**, or be aware of the need to void.
- **Evidence**:
 - A systematic review of RCTs in care homes showed fewer incontinent episodes.
 - PV is effective short term for **daytime UI** in nursing home residents and home care clients **when caregivers comply**.
 - **25–40% of incontinent nursing home residents respond well**, with daytime episodes falling to one or none. Over a third cannot toilet successfully even with help.
- A **3-day trial of PV** shows whether it is likely to work.
- An RCT in nursing homes had staff trained for at least 4 hours prompt residents every 2–2.5 hours. It reduced wet episodes and was **sustainable for 6 months**.
- **Functional incidental training (FIT)** combines PV with activities of daily living and restorative care (a walking programme and exercise), using a designated nursing assistant role. It repeatedly improves **both physical function and UI** in nursing home residents.
- **Ultrasound-assisted PV** may beat conventional PV. In an RCT in Japanese nursing homes, staff scanned bladder volume and prompted a void at optimal capacity, rather than every 2–3 hours. This produced a **greater reduction in daytime urine loss**. QoL was unchanged in both groups, and **caregiver burden worsened with conventional PV**.

TABLE 114.5 UNDERSTANDING TOILETING PROGRAMMES

Programme	Patient role	Schedule	Key components and suitability
Timed voiding (scheduled toileting)	Passive	Fixed, predetermined intervals	Institutional example: 8 times in 24 hours, about every 3 hours — on waking, after breakfast, mid-morning, before lunch, after an afternoon nap, before dinner and at bedtime; night-time waking only if the person wishes
Habit training	Passive	Fixed but individualised to the identified pattern of incontinence, to pre-empt it	Diary identifies incontinence and voiding patterns; in nursing homes the schedule is often built around daily events

Programme	Patient role	Schedule	Key components and suitability
Prompted voiding	Responsive: must request or accept toileting	Scheduled checks, as for timed voiding	Monitoring, prompting, praise. Most successful in those who can ask for help or void when prompted.
Functional incidental training	Responsive	As for prompted voiding	Prompted voiding plus low-intensity endurance and strength exercise , mobility and transfer training
Bladder training	Active	Established schedule with the interval increased incrementally	Requires ability and willingness to take part (independent toileting, or with help). Components: 1. Education on LUT physiology and pathophysiology — written, visual and verbal. 2. Scheduled voiding with systematic delay: voiding by timetable, not urge. 3. Urge suppression: quick PFM contractions, deep breathing with relaxation, distraction. 4. Gradual increase in the voiding interval. 5. Reinforcement through encouragement and positive feedback.

From Flanagan et al., 2012; Newman et al., 2014; Newman and Wein, 2009; Newman, Eason and Specht, 2018.

- **What the table tests:** the axis is **patient role** — TV and HT are done **to** the patient, PV **with** the patient, BT **by** the patient.
- **HT differs from TV only in individualising the schedule.**
- **Only BT aims to lengthen the interval;** the caregiver programmes aim only at dryness.

Challenges of Caregiver-Administered Voiding Schedules

- Toileting regimens impose a significant burden on caregivers. UI management is rated the **third most troublesome caregiving task**, after lack of time for one's own needs and managing the recipient's emotional and behavioural problems.
- Specific strains are odour, toileting, changing pads and clothing, the cost of pads and briefs, wet clothing, and changing bed linen.
- In **dementia at home**, prompting can cause irritation and arguments, because the person feels **treated like a child**.

Delayed Voiding

- Delayed voiding (DV) is another way to lengthen the interval between voids, but **without a fixed schedule**.
- **Protocol:**
 1. At the first urge, use urge suppression until the urge subsides.
 2. Do not then go straight to the toilet; **wait 5 minutes before voiding**.
 3. Once comfortable at 5 minutes, increase the wait step by step towards a normal interval.
- DV combines with BT, PFMT and urge suppression to reduce excessive frequency.
- Patients who have had UUI rush at even a mild urge for fear of leaking. Most can be persuaded to try a 5-minute delay **in a safe setting, such as at home alone**. They are often surprised that the urge fades, which builds a **sense of control and confidence**.

	Bladder training	Delayed voiding
Trigger to void	The clock (scheduled interval)	The urge, followed by a fixed wait
Schedule	Yes; interval raised 15–30 minutes a week	None; the wait starts at 5 minutes and rises gradually
Shared elements	Urge suppression; the goal of a normal voiding interval; patient-dependent	

Concept — *Scheduled voiding regimens lie on a spectrum from active (BT, DV) to passive (TV, HT). Where a patient sits on it is set by cognition and function, and the caregiver programmes succeed or fail on caregiver consistency rather than patient skill.*

BEHAVIOURAL TREATMENT FOR VOIDING AND PELVIC FLOOR DYSFUNCTION

- **PFM overactivity (high-tone PFM dysfunction)** is characterised by elevated resting tone (hypertonicity) and reduced ability to relax. It presents with:
 - pelvic floor tenderness or pain;
 - **bladder pain syndrome**;
 - voiding symptoms — **hesitancy, straining and incomplete emptying**.
- **Assessment:** digital palpation for tenderness and pain, resting tone, contraction strength and ability to relax. Some women have generally high resting tone; others tense **only when attempting to void**.
- **Treatment** teaches **conscious PFM relaxation, especially during voiding**. The first step is education on anatomy and function:
 - normal voiding is coordinated: **pelvic floor relaxation precedes, and can initiate, detrusor contraction and urethral relaxation**;
 - some women habitually void by **Valsalva**, believing they must push. This provokes a **reflex PFM contraction that impedes flow**;
 - patients must learn that straining is unnecessary and counterproductive, and that **voluntary PFM relaxation lets the bladder empty naturally**.
- PFMT here builds **awareness of tension as distinct from relaxation**. An active contraction shows what tension feels like, lets the patient contrast it with relaxation, and **leads to a more complete relaxation afterwards**. BF brings muscle tension to conscious awareness.

	Up-training	Down-training
Target	Contraction (strength, control)	Relaxation of a high-tone PFM
Difficulty	Easier to teach	Often harder to teach
Work-to-rest ratio	1:1 or 1:2	1:4 or longer , emphasising the relaxation phase
Aids	Palpation, BF, E-stim	BF-assisted PFMT , especially for pain patients, coordinating movement with the visual signal

- **Down-training sequence:** the patient learns to feel the muscle release and relax → then coordinates relaxation after a contraction → then carries the relaxation into voiding.
- **Generalising to voiding:** many women rush voiding because of busy lives. They are encouraged to create a relaxing environment and allow adequate time.
- **Good voiding technique:** slow down → take a deep breath → relax the body → relax the PFMs → **wait for the urine to flow**.

- **Double voiding** (anecdotal benefit): linger for a second detrusor contraction to empty more completely. Rise slightly from the seat, sit back down and relax the PFMs again.

Concept — *In high-tone dysfunction the same PFM skill set is **run in reverse**. Relaxation, not contraction, is the target, which is why the work-to-rest ratio flips from 1:1–1:2 to 1:4 and why straining makes voiding worse.*

LIFESTYLE MODIFICATIONS

- Lifestyle modifications change habits and lifestyle choices that **may contribute to UI and LUTS**. They involve a conscious choice to change a practice under the patient's control, and are sometimes called **self-care practices**.
- They can occasionally be **primary** but are **usually adjunctive**.
- They cover fluid management, reducing caffeine and other dietary irritants, managing constipation, smoking cessation and weight loss.
- The evidence is much thinner than for the skill-based treatments, yet these measures substantially affect bladder health and are integral to conservative management.

Fluid Management

- Fluid may help prevent several urinary diseases, and underhydration harms the urinary system. Changing fluid type or volume is used as a primary or adjunctive strategy.
- **Assessment:** a **24- to 48-hour intake and output diary**, with voided volumes where possible. It reveals excessive intake, inadequate intake, and diurnal patterns that contribute to LUTS.

	Excessive intake	Inadequate intake
Effect	Large volumes trigger urgency, frequency and UI. Very high intake (about 3.7 L a day, against about 2.4 L) is linked with more frequency and UI. In healthy young men, excess intake also raised bladder pressure on ambulatory urodynamics.	Risk of dehydration, especially in older women . It may contribute to urinary tract infection (UTI) , constipation and reduced functional capacity . Women who drank less than they wished had more UTIs.
Why	Deliberate "flushing" of the kidneys, weight loss, or habit. In the Boston Area Community Health (BACH) interviews, women cited health messages ("water is good for you"); men cited warding off infection, flushing the kidneys and keeping cool.	Self-imposed restriction to control symptoms , reported across all racial and ethnic groups in BACH, by community-dwelling elders and by working women
Frequency in practice	Less common	More common
Action	Reduce excess when output is abnormally high (above about 2100 mL in 24 hours). This helps urgency, frequency and UUI.	Increase towards the target

- **Target intake:** **six to eight 8-ounce glasses, or 30 mL/kg a day, with a minimum of 1500 mL a day**, at designated times, unless a medical condition contraindicates it.
- **Institute of Medicine (2005):** total water **2.7 L a day for women, including water in food**. This food component causes confusion, because fluid is often assumed to come only from beverages.
- An older adult who restricts fluid for fear of LUTS and is at risk of dehydration should keep an **oral fluid record**.
- **Persuasion tip:** concentrated urine smells stronger. Patients are sensitive about odour and welcome dilution as a way to reduce it.

Timing of Fluid Intake

- Overall restriction is not a good strategy, but **restriction timed to limited toilet access** (for example, before an outing) helps. The patient should make up the missed fluid earlier or later so that the daily total stays optimal.
- **Evening restriction** for nocturia: stop fluids **3–4 hours before bedtime**.
- **Daytime fluid retention mobilised during sleep** causes nocturia. Manage the daytime accumulation with **support stockings**, or **elevate the legs in the late afternoon** to mobilise the fluid well before bed.
- **Loop diuretics**:
 - A mid- to late-afternoon dose completes the diuresis before bedtime.
 - Loop diuretics aggravate UI by **speeding bladder filling and causing sudden urges**. Options are to stop the drug, **switch to a non-loop diuretic**, or change the timing.

Evidence for Fluid Management

- Fluid management is widely used, but the evidence is **limited and mixed**. A review found evidence that **reducing intake improved weekly UI episodes**.
- Two small RCTs:
 - In OAB, reducing intake by a quarter reduced daytime frequency, urgency and nocturia; increasing intake worsened frequency but not urgency, nocturia or UUI.
 - In women with urodynamically confirmed idiopathic DO, reducing intake reduced frequency and UUI episodes.
- By contrast, the **Nurses' Health Study**, a large prospective cohort followed for 4 years, found **no association between total fluid intake and incident UI** of any type.

Concept — *The aim of fluid management is a normal range, not restriction. Both excess and deficit worsen LUTS, and most patients seen in practice are over-restricting. Timing and type are adjusted before volume is cut.*

Caffeine Reduction

- Caffeine worsens urgency, frequency and UUI **partly as a diuretic and partly as a bladder irritant**. Intake from beverages, foods and medications is easily underestimated. Most US adults and most children consume caffeine; in children, coffee and energy drinks are rising while soda is falling.
- **Urodynamic effects**:
 - it **raises detrusor pressure during filling** and is a risk factor for DO in women with UI;
 - it **lowers the threshold of bladder sensation** in the filling phase, raises flow rate and voided volume, and causes diuresis (at 4.5 mg/kg in humans);
 - in mice, daily oral caffeine caused DO and increased bladder sensory signalling.
- **Epidemiology**:
 - Caffeine is associated with LUTS in both sexes. In the Nurses' Health Study II, high caffeine intake was associated with frequent UI.
 - In **BACH**, women who increased coffee intake had greater progression of urgency; recent increases in soda, especially **caffeinated diet soda**, were linked to worse scores, urgency and LUTS progression. In men, more coffee or caffeine at baseline predicted **storage symptom progression**.
 - **About 2 cups of coffee a day (250 mg)** is significantly associated with moderate-to-severe UI in US men.

- **Nocturia:** coffee is not associated with nocturia in either sex. Tea is, but only slightly, and only in women drinking **3 or more cups a day**. Women with nocturia were less likely to drink caffeine after 6 pm.
- **Intervention evidence:**
 - Reducing caffeine reduces episodes of **both SUI and UUI**.
 - A scoping review found observational associations with **urgency** but unclear associations with other LUTS.
 - Interventions reduced **urgency and nocturnal enuresis**; results for UI episodes and frequency were inconsistent.
- The **ICI recommends caffeine reduction**, but the amount needed is unknown. Sensitivity varies, and only a **subset of patients with urgency or UUI is caffeine sensitive**.
- **Practical approach:**
 1. Offer a **3- to 5-day trial** of reduction to test sensitivity.
 2. If symptoms improve, patients are usually more willing to cut down or stop.
 3. Reduce **gradually, mixing caffeinated with decaffeinated beverages over several weeks**, to avoid withdrawal headache and irritability.

Other Dietary Irritants

- Suspected irritants include **aspartame, citrus fruit, highly spiced foods and tomato products**. Many clinical cases improve when they are reduced, but this does **not** mean every patient with LUTS should eliminate them.
- Use a **food and beverage diary** to identify the patient's own irritants, then **eliminate them one at a time** to confirm the relationship. In an elimination diet, suspect items are withdrawn separately and in succession until the culprit is found.
- **Alcohol:**
 - It is thought to worsen urgency, frequency and UI through diuresis, but support is weak. In Japanese women the increase in UI risk was not significant.
 - In men, alcohol has been called an avoidable risk factor for LUTS, and high consumption is associated with moderate and severe LUTS. However, a prospective cohort found **beer protective against OAB**.
 - A review consistently found **modest drinking associated with fewer symptoms than abstinence**, and **heavy drinking with more UI and LUTS**.
 - Caffeinated alcoholic drinks need study.
- **Acidic fruit:** no relation to UI in the Nurses' Health Study.
- **Carbonated and artificially sweetened beverages (ASBs):**
 - Avoidance is widely advised on clinical experience rather than formal evidence.
 - In rats, artificial sweeteners **enhance detrusor contraction**.
 - In a prospective cohort of women, carbonated soft drinks were **independently associated with onset of OAB and SUI**.
 - In the **Women's Health Initiative Observational Study (WHI-OS)**, ASB consumption was associated with higher odds of UI, rising with intake.
 - Whether reducing ASBs relieves symptoms is untested.

Bowel Function

- **Chronic constipation (fewer than 3 stools a week)** and straining at stool are linked to LUTS. The ICI found "**some evidence**" that chronic straining at defecation is a risk factor for UI.

- **Why they are linked:**

- the bladder, urethra and rectum lie close together and share innervation, so reciprocal effects are likely;
- excessive straining is associated with **denervation of the external anal sphincter and PFM**s, and lifelong straining may have a cumulative effect;
- animal and clinical data support **bladder–bowel cross-sensitisation**.

- **Evidence:**

- In the **Epidemiology of LUTS (EpiLUTS) II** survey, OAB was more often reported by men and women with chronic constipation.
- In a case-control study, constipation and straining were more common in women with LUTS, including DO and urgency.
- In a prospective cohort, women with UI had **worse bowel function** (constipation and diarrhoea scores), and worse UI of any subtype went with worse faecal incontinence, diarrhoea and constipation.

- **Faecal impaction**, notably in nursing homes, can irritate the bladder (OAB) or **obstruct voiding**, causing incomplete emptying and **overflow incontinence**.

- **Management:**

- take a bowel history at the initial assessment;
- ensure adequate dietary **fibre and fluid** — **fibre does not work without sufficient fluid**;
- encourage regular exercise and use external stimulation;
- set a regular evacuation time, preferably **after a meal**, to use postprandial bowel motility.

Concept — *Bladder and bowel share space, nerves and pelvic floor, so dysfunction in one travels with the other. Constipation is therefore treated as part of the bladder programme, not as a separate problem.*

Obesity and Weight Reduction

- Obesity is **one of the most clearly established risk factors for UI**. Epidemiological and interventional data, including several systematic reviews, link body mass index (BMI) to UI and other LUTS. **Obese women have about double the risk of UI.**

- **Proposed mechanisms:**

- **chronically raised intra-abdominal pressure** on the bladder and pelvic floor;
- **metabolic syndrome** predisposing to UUI;
- obesity-related **inflammation and oxidative stress** altering collagen metabolism.

- **Interventional evidence:**

- UI improves in morbidly obese women after the dramatic weight loss of **bariatric surgery**. In national claims data, resolution of a UI diagnosis was more frequent after bariatric than after non-bariatric surgery, although the incidence of new UI did not differ.
- Intensive lifestyle therapy with a low-calorie diet reduced UI in **prediabetic women**.
- Modest weight reduction reduced **SUI episodes but not UUI**.
- **Loss of 5–10% of body weight**, sustained over 12 months, is enough to reduce incontinence episodes substantially.

- **Most definitive study:** an RCT of an intensive 6-month group weight-loss programme (diet, exercise and behaviour modification) against structured education, both with a self-help behavioural UI booklet. The weight-loss group lost more weight and had a **significantly greater reduction in UI episodes**.

- Clinicians should recommend **weight loss as part of lifestyle treatment for overweight women with UI**. Because BMI is modifiable, keeping weight within guidelines may also **prevent UI**.

Concept — *Obesity is a modifiable pressure load, and the evidence runs from association through to randomised intervention. A 5–10% loss is enough, which is why behavioural weight loss is the only grade A lifestyle recommendation.*

ADHERENCE TO CONSERVATIVE TREATMENT

- Effectiveness depends on an **involved, motivated patient**. Most behavioural interventions are forms of **self-management**, so the clinician's main challenge is to motivate the patient to take part, practise consistently and persist long enough for meaningful change.
- Progress is **gradual**, which makes compliance harder for patients expecting quick results.
- **Clinician support:**
 - set realistic expectations — improvement takes **weeks to months** and may be irregular, with good and bad days;
 - schedule follow-up to maintain accountability, track and reinforce progress, find and address barriers, adjust the regimen and encourage persistence;
 - reassure patients that they are not alone.
- **Measurement of adherence has often been overlooked in research.** Few studies report adherence to exercise, strategies or schedules, and methods for improving adherence are sparsely studied.
- A secondary analysis of **ATLAS** (see Vaginal Devices) found that adherence to PFME and bladder control strategies can be **high and sustained** when delivered by trained interventionists.
 - **Commonest barriers: remembering to exercise and finding time.**
 - **Remedies:** reminders (timers, text messages), exercise aids, linking exercise to environmental cues, and building it into daily activities.
- **Determinants of PFMT adherence:** positive intention to adhere, **self-efficacy**, attitudes to the exercise and its benefits, perceived social pressure, and **therapist enthusiasm**. Patients and clinicians agree that **patient-related factors** are the main barriers.
- A review of qualitative studies identified **six modifiers of adherence:** knowledge; physical skill; feelings about PFMT; cognitive analysis, planning and attention; prioritisation; and service provision.
- Clinicians should **assess adherence during and at the end of therapy** and use patient-focused strategies.
- **Long term:**
 - PFMT adherence **declines over time and varies widely**; positive intentions and self-efficacy predict long-term adherence.
 - Long-term outcome studies are few and inconsistent but promising, since many patients sustain their improvement.
 - More work is needed on why patients regress and how to maintain benefit.

Concept — *Because behavioural therapy is self-management, its efficacy is capped by adherence. The clinician's real intervention after teaching the skill is **sustaining motivation**, which is why supervised and intensive programmes outperform written instruction.*

PREVENTION OF URINARY INCONTINENCE

- Conservative interventions are non-invasive, which makes them suitable for prevention.

- Primary prevention with PFMT, behavioural training or lifestyle change (weight loss, fluid moderation, diet) has been studied in **three at-risk groups: older women, childbearing women, and men undergoing prostatectomy.**

Older Females

- **Age is perhaps the most established risk factor for UI**, yet prevention in older women has been little studied.
- **First prevention RCT:** a group behaviour modification programme against no intervention in continent postmenopausal women aged **55 or over**. At 12 months the programme gave better continence status, PFM strength (pressure and displacement scores), voiding frequency and intervoid interval.
- A feasibility study of a 6-week pelvic fitness and education programme in women aged 65 or over in independent living improved symptom bother, symptom distress (Urogenital Distress Inventory-6, UDI-6) and QoL (Incontinence Impact Questionnaire short form, IIQ-7).
- An RCT in women aged 55 or over with no or minimal UI compared a **2-hour bladder health class** with a **20-minute video**. Neither group progressed in UI or urgency over 12 and 24 months, and the two formats were comparable.

Childbearing Females

- **Pregnancy and childbirth are established risk factors**, and most prenatal classes teach the pelvic floor and PFM exercise.
- In **primiparous** women, **intensive antenatal PFMT** with one-to-one instruction to secure correct contraction and continued supervision prevents UI in **late pregnancy (35–36 weeks)** and in the **short-term postpartum period (6 weeks to 6 months)**.
- A large Cochrane review concluded that antenatal PFMT in **continent** pregnant women may reduce UI in late pregnancy and in the **mid-postnatal period (more than 3 to 6 months)**.
- The effect on **long-term UI and in multiparous women is less clear**.
- **ICI recommendations** are those in Table 114.2: the 12-week antenatal programme from 16–24 weeks, a postnatal programme after forceps or a baby of 4000 g or more, and education for older women.

Males Undergoing Prostatectomy

- **PPI is almost universal immediately after prostatectomy**. It usually improves over weeks to months, and the literature on behavioural treatment is mixed, but **PFMT can speed the return of continence**.

Timing	Rationale	Evidence
Preoperative	Men learn and practise PFM control before they need it	A single session of BF-assisted behavioural training shortened UI duration and reduced severity over the 6 months after RP
Immediately after catheter removal	Early training during recovery	Inconsistent, but generally beneficial at least in the first 3 months
Combined pre- and postoperative	Both	Reduced severity at 3 months , and in some studies up to 12 months

- **Practice gap:** most men undergoing RP still receive **no PFMT instruction**, and some clinicians wait and treat only persistent UI. Even transient UI is distressing, however, and hastening recovery matters for activity, social life and well-being.

Concept — *Prevention targets the three predictable insults — ageing, childbirth and prostatectomy. In each, the proven benefit is earlier or less incontinence in the short term rather than established long-term prevention.*

FUTURE OF LOWER URINARY TRACT SYMPTOMS PREVENTION AND BLADDER HEALTH PROMOTION RESEARCH

- Prevention of UI and LUTS is a growing emphasis. The ICI has examined **education-based interventions to raise public awareness** and promote population bladder health, and delivery models for skill-based prevention in at-risk groups (older adults, childbearing women, men undergoing RP).
- **2010**: a multidisciplinary expert panel issued a **consensus statement on the healthy bladder**. It aimed to raise awareness, encourage primary prevention through public education, reduce the personal, societal and economic impact of bladder conditions, and propose research-based clinical directives.
- **2015**: the NIDDK formed the **Prevention of Lower Urinary Tract Symptoms (PLUS) Research Consortium**, a **seven-site network** developing the concept of bladder health and foundational work to inform policy on preventing LUTS in **women across the life course**.

VAGINAL DEVICES FOR INCONTINENCE

- **Two types of mechanical device:**
 - **intravaginal devices** that **support the bladder neck and urethra**;
 - **intraurethral or meatal devices** that **block urine passage**.
- Devices are used mostly for SUI, but also for UUI and MUI.
- **No external meatal or intraurethral device for UI is currently available in the United States.**
- A 2011 Cochrane review found **insufficient evidence** to recommend any specific device, or to show that devices beat other treatments.
- The **ideal device** controls leakage adequately, is easy to apply or insert, has few adverse effects and is cheap.

Intravaginal Devices

- **Rationale:** when pelvic support is weak, the urethra drops to a position where a rise in abdominal pressure **cannot be transmitted to it** to aid closure. An intravaginal device **restores the upper urethra above the pelvic floor**, where abdominal pressure again helps close it. A continence pessary achieves this.
- The vaginal pessary is one of the **oldest medical devices**, used for centuries for POP and more recently for SUI. The **Dutch College of General Practitioners** recommends offering women with SUI a **choice of PFMT or a pessary** as initial treatment.
- **How a continence pessary works:**
 - it stabilises and supports the urethra and bladder neck;
 - it **increases urethral length**;
 - it gently **compresses the urethra against the pubic bone** when abdominal pressure rises.
- The **knob of an incontinence ring sits against the urethrovesical junction** (Fig. 114.13) and supports it much as a surgically placed vaginal sling does.
- Women with urgency and frequency caused by pressure from POP may also benefit.
- **Materials:** inert plastic or silicone, which prevents odour and absorption of secretions. Designs specific to SUI are the **Hodge (Fig. 114.14)**, **dish (Fig. 114.15)** and **ring (Fig. 114.16)**, each available without or with support.
- **Contraindications (few):** **active pelvic or vaginal infection; severe ulceration; allergy to silicone or latex; likely non-compliance with maintenance and follow-up.**
- Difficulty with self-removal and insertion limits wider use.

- **Side effects:**

- common: **vaginal discharge and odour**;
- rare but serious: **vesicovaginal fistula, rectovaginal fistula, erosion and impaction**;
- the severe complications followed **long-term use (6–10 years) for POP** and included bleeding, severe discharge, pain and constipation, often more than one type.
- A continence pessary can be worn **continuously or intermittently**, for example only during exercise.
- **Evidence is limited.** SUI and MUI improved with a pessary in about half of women in two trials.
- **ATLAS** (Ambulatory Treatments for Leakage Associated with Stress Incontinence; National Institute of Child Health and Human Development [NICHD] Pelvic Floor Disorders Network) randomised women with **stress-predominant UI** to pessary, behavioural therapy, or both.
 - **At 3 months:** "much better" or "very much better" in 40% with pessary and 49% with behavioural therapy — **not significantly different**.
 - Behavioural therapy was significantly better for **no bothersome symptoms** (49% versus 33%) and **satisfaction** (75% versus 63%).
 - **Combined therapy** was better than pessary alone but **not better than behavioural therapy alone**.
 - **At 12 months:** satisfaction exceeded 50% in all groups, but **no difference was sustained** on any measure.
- For **OAB**, a retrospective cohort found **similar cure with a ring pessary and multicomponent behavioural therapy**.

INTRAVAGINAL CONTINENCE DEVICES COMPARED

	Continence pessary (ring, dish, Hodge)	Uresta	Impressa
Design	Reusable inert silicone or plastic; knob or support against the urethrovesical junction	Bell-shaped pessary with a handle at its base; narrow tip	Disposable, single-use, tampon-like: silicone core with flexible anchor and resin support poles, a soft nylon mesh cover, and an applicator
Fitting	Fitted by a clinician	Three sizes (3, 4, 5), fitted by a clinician or by the patient	Inserted by the patient through an applicator; removed by string
Mechanism	Supports and lengthens the urethra; compresses it against the pubis	Self-positions so that the wide base supports the urethra	Anchors without migrating. The mesh stretched between the poles forms a tension-free suburethral sling whenever abdominal pressure reaches the pelvic floor, without obstructing urine or secretions.
Wear	Continuously or intermittently	Inserted like a tampon	Non-absorbent. Need not be removed to void or defecate; remove during menstruation and intercourse.
Evidence	About half improve; ATLAS as above	Small study: fewer UI episodes and lower pad weight, no complications; most satisfied at 2 weeks; half still using it at 12 months, with significant benefit	Open-label study in urodynamically proven severe SUI: most achieved over 70% reduction in pad-weight gain. Adverse events were discomfort, pain, spotting and one candidiasis, with no mucosal injury. RCT against a continence pessary: high success with both at 4 weeks on PGI-I, no difference, high attrition.

Figs. 114.13–114.18 show the incontinence ring with its knob at the urethrovesical junction, the Hodge, dish and ring pessaries each without and with support, the Uresta in place, and the Impressa in the vagina.

Concept — *Every intravaginal continence device does one thing: it restores suburethral support so that abdominal pressure closes the urethra rather than displacing it. The devices differ only in delivery — fitted and reusable, self-fitted, or disposable — and none has shown superiority over behavioural therapy.*

MODELS OF DELIVERY

- Despite the evidence and guideline endorsement, conservative interventions are **poorly integrated into practice and not widely accessible**, chiefly because there are too few qualified providers.

Integration of Continence Specialists in Clinical Practice

- A successful model is to integrate **APPs — continence nurse practitioners and physician assistants** — into urological practice. APPs have historically embraced behavioural treatment as a specialty for non-surgical management of many LUT conditions.
- Fig. 114.19 gives an algorithm for a behavioural treatment pathway for LUTs; its legend defines post-void residuals (PVRs).
- Urology suits a multidisciplinary **Continence and Pelvic Health service** that offers comprehensive surgical and medical care from experts in UI, voiding and pelvic floor dysfunction.
- Changes in reimbursement for BF-assisted PFMT, E-stim and **posterior tibial nerve stimulation** have let urologists extend beyond drugs and surgery to multifaceted behavioural programmes.

Group Administration Models

- PFMT has long been part of **prenatal education and fitness classes** for pelvic health in pregnancy and postpartum. It has also been built into general fitness programmes and specific pelvic health classes to **treat** symptomatic women. These focus on strengthening and need **multiple sessions over several weeks**.
- **Bladder health classes** in a classroom or community use a broader health-class format: anatomy and function, voiding habits, toileting technique and bladder control strategies. They are effective both for **preventing symptoms in older populations** and for **reducing existing symptoms**.
- **Group-Administered Behavioural Treatment (GBT) trial**: a multisite RCT in community-dwelling women with UI compared a **one-time 2-hour bladder health class with a PFME audio CD** against no treatment.
 - ICIQ-SF scores were consistently lower with GBT through 12 months, but the difference fell short of the projected 3 points.
 - Conclusion: GBT is **modestly effective and cost-effective** in older women.
- Such programmes are less effective than individually supervised therapy, but they are **risk free, cheap and deliverable in medical or non-medical settings**. They reach people with mild or preclinical symptoms, teaching treatment, prevention and delay of onset, and signposting conventional care.

Self-Administration Models and Mobile Health

- Pamphlets on PFMT, BT and lifestyle change served as stand-alone instruction or as adjuncts to drugs or clinic treatment. Some grew into comprehensive step-by-step programmes, and technology has moved them to telehealth devices and websites.
- **Internet versus postal self-training for SUI**: an RCT compared an internet programme (email support from a urotherapist plus cognitive behavioural assignments) with a postal programme.
 - Both gave large improvements, with **no difference in primary outcomes**.
 - The internet group did better on several secondary outcomes.

- Gains were maintained at **1 and 2 years**.
- **Mobile health (mHealth) apps:** the commonest teach PFMT for SUI; others target pregnancy and postpartum. Free and paid apps are of **very variable quality**. On the validated Mobile App Rating Scale (MARS), many were poor and **only one was supported by scientific evidence**.
- **Evidence-based apps:**
 - The **Tāt** app for SUI (three sets a day of slow, quick and pre-cough contractions) improved ICIQ-SF by about the **minimal important difference** in an RCT. An updated version was effective for **UUI and MUI**.
 - Comprehensive programmes add urge suppression and lifestyle content:
 - an electronic tutorial with video vignettes on using the PFMs for urge triggers and nighttime waking;
 - **MyHealtheBladder** for female veterans, an 8-week programme of daily modules on PFME, bladder control, fluid management, risk-factor reduction and self-monitoring, which reduced ICIQ-SF;
 - a similar programme for **perioperative behavioural training in men undergoing RP**.
- **Reviews:**
 - App-based PFMT for SUI "shows promise" for efficacy and adherence.
 - A scoping review found **level 2 evidence** of benefit in symptoms, satisfaction, adherence and cost, with good fidelity to optimal PFMT parameters.
- mHealth improves **access** and may improve **motivation and adherence**. Useful features are BF linkage, audio instruction, video, stories and testimonials, and reminders such as text messages. Research is now focused on barriers, facilitators and features that improve uptake, adherence and outcome.

Concept — *The constraint on conservative therapy is **access, not efficacy**. Each delivery model — specialist APPs, groups, self-administration and apps — trades **intensity, and therefore effect size, for reach**.*

KEY POINTS

- Conservative behavioural treatments improve UI and other LUTS by **changing behaviour or environment, or by teaching new skills**.
- There is **strong evidence** that conservative interventions are effective **first-line treatments** for UI and other LUTS.
- Effectiveness relies on the **active participation of an involved, motivated patient**.
- **Intensive** behavioural interventions involving an **experienced clinician** are more effective.